

MAINTENANCE OPERATIONS CLIENT

Data Cleaning & Preparation Methodology

Maintenance Operations Analytics & Reporting Automation

Document Control	Detail
Engagement	Aug-Nov 2024
Role	Business Analyst + Data Analyst
Primary Stakeholder	Maintenance Manager
Data Scope	Approximately 10,000 historical records across 20 teams
Confirmed Sources	Excel, CSV, Google Sheets and database
Tools Used	Excel, SQL, Python, Power Query, Google Sheets and Power BI
Status	Portfolio methodology reconstructed from confirmed project facts

Integrity note: The project definitely involved cleansing and structuring the maintenance data and the listed quality issues. Exact historical scripts, transformation order, thresholds and source-specific rules are not retained. Detailed steps below are therefore labeled as a reproducible portfolio methodology rather than exact historical ETL code.

1. Purpose

The purpose of data preparation was to convert fragmented, inconsistent historical maintenance records into a controlled analytical dataset suitable for operational KPI calculation and Power BI reporting.

Confirmed starting condition: Approximately 10,000 historical records across 20 teams were available through Excel, CSV, Google Sheets and database sources. The data was not analysis-ready.

2. Confirmed Data Quality Issues

Issue	Why It Matters
Duplicate records	Can inflate task/work-order counts and distort completion, backlog and performance measures.
Missing values	Can prevent correct classification, timing analysis or team/technician attribution.
Inconsistent team names	Can split one logical team into multiple reporting categories.
Inconsistent date formats	Can invalidate overdue and completion-time calculations.
Inconsistent status names	Can distort Completed/Open/Pending and overdue reporting.
Blank records	Can introduce non-actionable rows into operational aggregates.

3. Preparation Principles

- Preserve source meaning before standardizing values.
- Do not manufacture missing operational facts.
- Separate raw/source data from prepared analytical data.
- Apply business-rule-sensitive cleaning before KPI calculation.
- Make reconstructed transformations reproducible and testable.
- Retain unknown values explicitly where evidence-based correction is not possible.
- Protect confidential client, technician, asset and location information in public portfolio outputs.

4. End-to-End Preparation Flow

Portfolio methodology: Source intake -> profiling -> structural standardization -> duplicate control -> missing-value handling -> team/status/category normalization -> date validation -> field-level validation -> business-rule readiness -> reconciliation -> Power BI-ready analytical dataset.

This sequence is a portfolio reconstruction of a defensible preparation workflow. It should not be described as the exact historical execution order unless additional evidence becomes available.

5. Step 1 — Source Intake & Inventory

Activity	Portfolio Method
Source identification	Register Excel, CSV, Google Sheets and database inputs used for the analytical scope.
Scope check	Confirm the prepared dataset represents approximately 10,000 historical records across 20 teams.
Field inventory	Identify the confirmed analytical fields: Task/Work Order ID, Team, Technician, Asset/Equipment, Created Date, Due Date, Completion Date, Status, Priority, Category, Location and Remarks.
Raw preservation	Keep an unchanged source/raw layer or equivalent copy before transformations.
Confidentiality	Exclude or sanitize client-identifying and sensitive operational values from public portfolio data.

6. Step 2 — Data Profiling

Before changing values, profile the data to understand the extent and location of quality problems.

Profile Area	Checks
Volume	Record count by source and combined dataset.
Completeness	Blank/null profile for analytically important fields.
Uniqueness	Duplicate/candidate-duplicate profile, especially around work-order/task identity.
Categorical consistency	Distinct Team, Status, Priority and Category values.
Dates	Parse success, missing dates and suspicious chronology.
Attribution	Missing Team, Technician, Asset/Equipment or Location values.
Text fields	Blank or potentially sensitive Remarks values.

Evidence boundary: Exact historical profiling counts are not retained; the portfolio should not invent them.

7. Step 3 — Structural Standardization

Area	Reconstructed Preparation Rule
Column naming	Use consistent business-friendly names aligned to the Data Dictionary.
Text values	Trim unnecessary leading/trailing whitespace and normalize formatting where it does not alter meaning.
Date fields	Convert Created Date, Due Date and Completion Date to valid analytical date/datetime types.
Categorical fields	Prepare Team, Status, Priority and Category for controlled mapping.
Blank rows	Identify rows without actionable maintenance information and control them before aggregation.
Data types	Use portfolio analytical types; do not claim they are the original database types.

8. Step 4 — Duplicate Control

Confirmed issue: Duplicate records existed in the source data.

The exact historical duplicate-identification algorithm is not retained. The portfolio rebuild should therefore use a documented, evidence-based approach rather than assuming every repeated identifier is a duplicate.

Control	Method
Exact duplicate detection	Identify rows that repeat the same relevant values across the selected comparison fields.
Candidate duplicate review	Investigate repeated Task/Work Order IDs or highly similar records.
Business distinction	Separate true duplicates from legitimate repeated maintenance activity.
Resolution	Remove, consolidate or retain records according to the documented reconstruction rule.
Validation	Compare pre/post-cleaning counts and verify KPI counts are not inflated by duplicate rows.

Important: Task / Work Order ID is a confirmed field, but its exact historical primary-key/uniqueness constraint is not retained.

9. Step 5 — Missing-Value Handling

Field Group	Portfolio Handling Principle
Task / Work Order ID	Investigate blanks because reliable record-level counting may be affected.
Team / Technician	Preserve Unknown/Missing where no evidence-based attribution exists.
Dates	Do not fabricate Created, Due or Completion dates.
Status	Do not infer a valid operational status without evidence; retain an unmapped/unknown treatment where necessary.
Priority / Category	Preserve missing/unknown values unless supported by a source mapping.
Asset / Equipment / Location	Do not invent attribution.
Remarks	No synthetic text is required for blank remarks.

10. Step 6 — Team Standardization

Confirmed issue: Team names were inconsistent.

Action	Reconstructed Control
Profile	List distinct team labels and frequencies.
Normalize	Remove harmless formatting inconsistencies such as whitespace/case where appropriate.
Map	Use a controlled mapping only where labels can be confidently linked to the same logical team.
Preserve ambiguity	Do not merge labels when equivalence cannot be supported.
Validate	Confirm standardized Team supports analysis across the project scope of 20 teams without inventing team names.

11. Step 7 — Status Standardization

Why status is critical: Status directly affects Completion Rate, Completed/Open/Pending reporting and the confirmed Overdue rule.

Action	Reconstructed Control
Profile values	Review distinct source status labels.
Normalize formatting	Control trivial spelling/case/whitespace variation where meaning is unchanged.
Map categories	Create documented analytical categories such as Completed, Open and Pending only where source meaning supports the mapping.
Unknown status	Retain as unknown/unmapped rather than forcing an unsupported classification.
Validate completion	Ensure records counted as Completed follow the same standardized category used in Completion Rate.
Validate overdue	Ensure Status != Completed is applied consistently with the confirmed overdue rule.

Historical boundary: The exact original status-value list and mapping table are not retained.

12. Step 8 — Date Standardization & Validation

Field	Preparation / Validation
Created Date	Parse to a valid date/datetime. Retain missing values rather than inventing them.
Due Date	Parse to a valid date/datetime; required for overdue classification.
Completion Date	Parse to a valid date/datetime for completed-work timing analysis.
Format consistency	Convert inconsistent source date formats into a common analytical representation.
Chronology	Investigate impossible or negative date intervals.
Invalid dates	Flag/control records that cannot be reliably parsed.

Overdue dependency: A record can be classified as overdue only when Due Date is valid, Due Date < Current Date and Status != Completed.

13. Step 9 — Priority, Category, Asset & Location Preparation

Field	Reconstructed Control
Priority	Standardize supported labels; preserve unknown/missing values.
Category	Standardize labels sufficiently for recurring-category analysis.
Asset / Equipment	Normalize only where supported; preserve meaningful identifiers/labels in the private analytical layer.
Location	Standardize only to the level supported by the source and confidentiality requirements.
Remarks	Use as contextual free text; sanitize sensitive information in public portfolio outputs.

14. Step 10 — KPI Readiness Checks

KPI / Analysis	Readiness Requirement
Completion Rate	Completed classification and eligible assigned-task population must be consistent; duplicates controlled.
Completed/Open/Pending	Status mapping documented and tested.
Overdue	Valid Due Date plus confirmed Status != Completed rule.
Average Completion Time	Valid dates and a documented reconstructed start/reference-date convention.
Priority Breakdown	Priority values standardized sufficiently for grouping.
Team-wise Completion	Team labels standardized before comparison.
Technician Performance	Technician attribution retained/handled transparently.
Backlog	Explicit reconstructed backlog definition documented before calculation.
Recurring Categories	Category values standardized.
Asset/Equipment Analysis	Asset/equipment attribution suitable for aggregation.

15. Step 11 — Reconciliation & Validation

- Compare record counts between raw, combined and prepared layers.
- Reconcile records removed or controlled as duplicates/blank records.
- Review missing-value counts in critical fields before and after preparation.
- Verify standardized team categories do not create unsupported new teams.
- Verify standardized statuses support consistent Completed/Open/Pending analysis.
- Test the confirmed Completion Rate formula against prepared data.
- Test that every overdue record satisfies Due Date < Current Date AND Status != Completed.
- Check that completed records do not appear as overdue under the confirmed rule.
- Validate date-based measures only on eligible valid records.
- Document any portfolio reconstruction rule that was not retained from the historical implementation.

16. Step 12 — Power BI Preparation

Once data-quality and KPI-readiness checks pass, the prepared dataset can feed the reconstructed Power BI semantic model.

Preparation Area	Requirement
Field naming	Align with Data Dictionary terminology.
Categorical values	Use controlled Team/Status/Priority/Category representations.
Dates	Expose valid analytical date fields required by reporting.
Measures	Implement KPI logic from the KPI Catalogue rather than redefining measures ad hoc.
Model design	Treat exact table structure/relationships as portfolio reconstruction unless historical evidence is supplied.
Confidentiality	Use sanitized/synthetic public values and remove sensitive free-text or identifiers.

17. Tool Responsibilities

The following tools are confirmed as part of the engagement. The exact historical division of every transformation across tools is not retained, so the portfolio should avoid claiming a specific step was executed in a specific tool unless reconstructed deliberately.

Tool	Safe Project Positioning
Excel	Source handling, review and analytical support.
Google Sheets	Source/working data environment.
SQL	Data querying and analytical support.
Python	Data preparation and analytical support.
Power Query	Data transformation and Power BI preparation.
DAX	Analytical measures in Power BI.
Power BI	Management dashboard/reporting solution.

Do not fabricate: Exact SQL queries, Python scripts, Power Query M code and historical DAX expressions are not retained.

18. Data Quality Control Matrix

Quality Dimension	Risk	Control	Downstream Impact
Uniqueness	Duplicate records	Duplicate detection/review	Counts, completion rate, backlog
Completeness	Missing values	Transparent field-specific handling	Attribution and timing analysis
Consistency	Team/status naming	Controlled normalization/mapping	Team/status KPIs
Validity	Date parsing	Type conversion and validation	Overdue/completion time
Accuracy	Unsupported inferred values	No fabrication principle	All reporting
Confidentiality	Sensitive operational data	Sanitization before publication	Portfolio safety

19. Preparation Acceptance Criteria

ID	Acceptance Criterion
DPC-01	Prepared data is structurally usable for Power BI reporting.
DPC-02	Duplicate records are identified and controlled using a documented reconstruction rule.
DPC-03	Missing values are handled transparently without fabricating operational facts.
DPC-04	Team and status inconsistencies are standardized sufficiently for governed KPI reporting.
DPC-05	Date fields required by overdue/timing analysis are parsed and validated.
DPC-06	Completion Rate can be calculated using the confirmed business rule.
DPC-07	Overdue classification can be tested against the confirmed rule.
DPC-08	Prepared dimensions support team, technician, priority, category and asset/equipment analysis where valid.
DPC-09	Portfolio outputs do not expose confidential client or sensitive operational data.
DPC-10	Reconstructed technical rules are clearly separated from retained historical facts.

20. Traceability

Preparation Area	Data Dictionary	KPI / Requirement Impact
Duplicate control	Task / Work Order ID	All record counts and rates
Team standardization	Team	Team-wise completion, workload imbalance
Technician handling	Technician	Technician performance
Date preparation	Created/Due/Completion Date	Overdue, completion time
Status mapping	Status	Completion, open, pending, overdue, backlog
Priority preparation	Priority	Priority breakdown, high-priority pending
Category preparation	Category	Recurring categories
Asset preparation	Asset / Equipment	Asset/equipment analysis
Location handling	Location	Optional location analysis
Remarks sanitization	Remarks	Contextual analysis and confidentiality

21. Portfolio Reconstruction Boundaries

Not Retained	Required Portfolio Treatment
Exact historical cleansing sequence	Use the documented reproducible methodology; label it reconstructed.
Exact duplicate algorithm	Define and test a portfolio rule.
Exact missing-value thresholds	Do not invent thresholds.
Original team/status mapping tables	Create sanitized mappings only when rebuilding.
Exact source schema/data types	Use the Data Dictionary's reconstructed analytical types.
Exact SQL/Python/Power Query code	Create new reproducible code and label it reconstructed.
Exact pre/post cleansing counts	Do not fabricate counts.
Exact refresh process	Do not claim a refresh method unless evidence is provided.

22. Methodology Summary

Preparation chain: Raw multi-source maintenance data -> profiling -> controlled cleansing and standardization -> KPI readiness -> reconciliation -> Power BI-ready analytical data.

This methodology must remain consistent with the Data Dictionary and KPI Catalogue. Later SQL, Python, Power Query and DAX artifacts should implement these controls rather than introducing contradictory cleansing rules.